

Sheaves Theory

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Goal of This Presentation

- 1 To understand the object of Presheaf, Stalk, Germs and Sheaf, and their morfism,
- 2 To build intuition behind those objects,
- 3 To understand the application of category theory.

Some Prelim: Category Theory

■ Category and Morphism definition

Definition (Category)

Define $\mathcal{C}$ as category if it has **object**, **morphism**, and **composition**, such that

- A class of $Ob\mathcal{C}$, called as class of object,
- For each pair of object, a set denote $\mathcal{C}(X, Y)$ or $Hom(X, Y)$ called as morphism, from X to Y ,
- For each triple (X, Y, Z) objects, we have a map $\mathcal{C}(X, Y) \times \mathcal{C}(Y, Z) \rightarrow \mathcal{C}(X, Z)$ define as composition, such that it has associativity and identity, such that
 - 1 If $f \in \mathcal{C}(X, Y)$, $g \in \mathcal{C}(Y, Z)$, and $h \in \mathcal{C}(Z, W)$, we have $h(gf) = (hg)f$
 - 2 For each object $X \in \mathcal{C}$, we have $id_X \in \mathcal{C}(X, X)$, such that for every $f \in \mathcal{C}(X, Y)$ and $g \in \mathcal{C}(Z, X)$, we have $f id_X = f$ and $id_X g = g$.

- Examples: Sets with functions, Group with group action, Poset with $\leq$
- Not a category example: the category of all category(?)

Some Prelim: Category Theory

- Functors
- Yoneda Lemma
- Examples that will be used
 - *Sets*
 - Ring of differentiable function $\mathcal{O}$

Some Prelim: Category Theory

- Colimit
- Adjoint

Some Prelim: Topology

■ Topological space definition

Definition (Topology)

Let X be a set. Define $\mathcal{T}$ as **topology** on set X if

- 1 $\emptyset$ and X in $\mathcal{T}$
- 2 The union of the elements of any subcollection of $\mathcal{T}$ is in $\mathcal{T}$
- 3 The intersection of the elements of any finite subcollection of $\mathcal{T}$ is in $\mathcal{T}$

We call X as **topological space** if it has $\mathcal{T}$ as its topology.

■ Open set definition

Motivation

- Why study sheaves?
- Because we need to use sheaves of function to explain schemes.
- Sheaf called as 'sheaf' because it glues every local data information. Presheaf is 'almost' sheaf, and it is still good idea to introduce it first.
- Fun Fact: Sheaves was invented by Jean Leray (a French Mathematician) in prison during WWII!

Restriction Map

- Let X be topological space. Suppose we have bunch of open sets, by let $U \subseteq X$ denote any open set and bunch of differentiable function on open set U .
- Those differentiable functions on open set U exudes a ring, denote this as $\mathcal{O}(U)$
- Notice if we have open set $V \subseteq U$, we can have a differentiable function at V .
- In another word, if $V \subseteq U$ such that we have inclusion map from V to U , we have a restriction map $res_{U,V} : \mathcal{O}(U) \rightarrow \mathcal{O}(V)$

- Now notice if we restrict a restricted differentiable function, we still have differentiable function over there.
- In another word, if we have $W \hookrightarrow V \hookrightarrow U$, we can have such diagram commutes.

$$\begin{array}{ccc} \mathcal{O}(U) & \xrightarrow{\quad res(U,V) \quad} & \mathcal{O}(V) \\ & \searrow \text{dashed } \exists res(U,W) & \swarrow res(V,W) \\ & \mathcal{O}(W) & \end{array}$$

Sheaves Intuition

- Now, let U be 'big' open set and $\{U_i\}$ denote open covers of U , such that $\bigcup U_i = U$.
- Suppose we have f_1 and f_2 differentiable function on U .
- If for every U_i , we have f_1 equal to f_2 on that U_i , then $f_1 = f_2$ to begin with.
- In another words, $\forall \{U_i\}_{i \in I}$ and $\forall f_1, f_2 \in \mathcal{O}(U)$, we have

$$\text{res}_{U, U_i} f_1 = \text{res}_{U, U_i} f_2 \implies f_1 = f_2$$

- This notion is called as **locality**. We will formalize this on sheaf definition.

- Next, with the same U and $\{U_i\}$, suppose we have f_1 on U_1, f_2 on U_2 and so on, and assume they equal on the pairwise overlaps.
- Notice we can just "glued" them together into one differentiable function on U .
- Or in another words, $\forall i, f_i \in U_i$ such that $\forall i, j, res_{U_i \cap U_j} f_i = res_{U_i \cap U_j} f_j$, then

$$\exists f \in \mathcal{O}(U) \forall i, res_{U, U_i} f = f_i$$

- This idea is called as **glueing**. Again, we will formalize this on the definition.
- Notice that this will work anyways with continous function, smooth function, and just plain function. Hence, this encourages us to define such nice properties formally.

Stalk and Germ Intuition

- Similar to previous slide's idea, can we find the intersection where the functions are equal?
- Then we define a relation such that

$$(f, U) \sim (g, V) \iff \exists W \subset U \cap V, \text{res}_{U,W} f = \text{res}_{V,W} g$$

for open set U, V and $f \in \mathcal{O}(U)$ and $g \in \mathcal{O}(V)$

- One can prove this is an equivalence relation.
- Then we can define a class such that,

$$\{(f, U) : p \in U, f \in \mathcal{O}(U)\}$$

for a point $p \in X$.

- This class is known as **germs**. We will define germs using this later.

- Consider a set of germs. Notice if we have two germs and add them together, you get another germs.
- Or explicitly speaking, let $f \in \mathcal{O}(U)$ and $g \in \mathcal{V}$, then $f + g$ is well-defined on $U + V$.
- To sketch the proof of well-defineness, suppose we have $\tilde{f}$ has the same germs to f , and also $\tilde{g}$ to g ,

Definition

- We will define presheaf in category theorist view.

Definition (Presheaf)

Define $\mathcal{F}$ as presheaf of a category $\mathcal{C}$ (can be a set, group, ring, top ...) over a category $\mathcal{D}$ if it is a functor such that

$$\mathcal{F} : \mathcal{D}^{op} \rightarrow \mathcal{C} \text{ (or } \mathcal{S}ets, \mathcal{G}rp, \mathcal{R}ing, \mathcal{T}op)$$

Or a contravariant functor from $\mathcal{D}$ to $\mathcal{C}$.

- For the scope of schemes theory, we let $\mathcal{D}$ be something of topological space.

Definition

Definition (Presheaf over Topological Space)

Let X be a topological space, define $\mathcal{F}$ as **presheaf over X** if it is a contravariant functor from $Open(X)$ (the category of open subsets of X) to a category $\mathcal{C}$. We can write it as

$$\mathcal{F} : Open(X)^{op} \rightarrow \mathcal{C}$$

- The category of $Open(X)$ has
 - Obj = open subset of U of X
 - Mor = inclusion $V \subseteq U$
- Of course one can check whether this is a category or not.

- This means that presheaf has following data (as a functor)

$$\begin{aligned} \forall \text{open set } U \subseteq X, U &\mapsto \mathcal{F}(U) \text{ (as a set or group or ring } \dots) \\ (V \subseteq U) &\mapsto (\mathcal{F}(U) \xrightarrow{\text{res}_{U,V}} \mathcal{F}(V)) \end{aligned}$$

- The identity and composition is obvious in previous explanation about restriction map.
- This explains why we define presheaf as contravariant.

Example

- The 'stalk' means a plant stalk, while the 'germ' means as 'cereal germ' to a stalk, as it is the heart of a function.

- Locality and
- Hence, presheaf failed as being 'sheaf' since it does not fulfill the locality (Uniqueness) and glueing(Existence).

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